# Sunanditha B S

Linux | Embedded Security | Automotive Cybersecurity

sunanditha24@gmail.com | Portfolio | LinkedIn | GitHub

## Education

**RNS Institute of Technology, Bengaluru** — CGPA: 8.95/10

*B.E. Information Science & Engineering* — *Expected 2027*

## Technical Skills

**Languages:** C, C++, Python, Bash, SQL **Systems & Tools:** Linux, Git, Docker **Automotive:** CAN, SocketCAN, UDS (ISO 14229), AUTOSAR **Security:** X.509, RSA-PSS, Secure Boot, OTA

## Projects

**BOLT: Boot & OTA Lockdown Toolkit** — 2026

*Python | RSA-PSS/SHA-256 | X.509 | Uptane-inspired OTA | pytest*

- Designed a secure OTA and boot integrity architecture for automotive ECUs, using a 2-tier X.509 trust chain and A/B partitioning to survive a crash mid-flash.
- Red-teamed the design against 4 attack classes (rollback/replay, signature stripping, MITM, power-loss truncation); all cases mitigated under test, verified with a 33-test pytest suite.

**CANary: CAN Bus Intrusion Detection Framework** — 2026

*Python | C | SocketCAN | ctypes | pytest*

- Built an 8-module CAN bus IDS integrating a native C-based 15-bit CRC checksum into a Python architecture via ctypes, tested across 3+ simulated ECUs.
- Wrote 6 rule-based detectors for spoofing, replay, flooding, and malformed frames with 4-tier severity alerts; 47 tests pass on live SocketCAN.

**WARDEN: UDS Diagnostic Session Security Simulator** — 2026

*Python | SocketCAN | UDS (ISO 14229) | pytest*

- Built a UDS (ISO 14229) session simulator over SocketCAN with seed-key SecurityAccess and attempt lockout, then wrote an attacker client that brute-forces the key.
- Detector flags repeated key attempts inside a sliding window by tracking session-state transitions and timing; 9 tests cover session logic and malformed input.

## Research

**EXCALIBUR: A Lifecycle Threat Model for Adaptive CAN Intrusion Detection** — 2026

- Found that the calibration phase of adaptive CAN IDS is an unguarded attack surface, then built a bounded-budget timestamp-drift attack that exploits it and validated it on real automotive CAN traffic (ORNL ROAD dataset) across 3 CAN IDs and 4 drive sessions.
- Designed and evaluated a median/MAD-based mitigation that measurably suppresses the attack.

**BICE: Behavioral Identity Continuity Engine** — 2026

- Built a per-device drift-based anomaly detector for IoT telemetry using Welford online statistics, catching 4 in 5 attacks with zero false positives across 9 devices, outperforming Isolation Forest and One-Class SVM baselines.

## Systems & Security Engineering

- Diagnosed and resolved two Linux boot failures via live-USB chroot recovery, involving a corrupt GPU driver and an out-of-order Intel VMD/RST controller.
- Rebuilt disk partitions at the sector level to relocate immovable NTFS $MFT mirror metadata, achieving a 60% capacity increase with zero data loss.
- Reverse-engineered a MediaTek boot trust chain (BROM → Preloader → LK → AVB) to load a custom system image on a locked bootloader.

## Achievements

TryHackMe Top 4% | Winner, Escape and Exploit Cybersecurity Competition | Top 15, ZeroDay Arena CTF